



AMBITION COLLEGE

Mid-Baneshwor, Kathmandu, Nepal

First Term Examination-2082

Subject : Web Technology(Subjective)

BCA 3th Semester

Time: 3 : 00 hours

Full Marks :- 50

Pass Marks :- 20

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group B

Attempt any SIX questions. [6 × 5 = 30]

1. Compare and Contrast between a block-level element and an inline element in HTML?
2. What is html ? Describe the structure of HTML
3. What is the strength of CSS? What are the various approaches(types) to include CSS documents in HTML ?
4. What is the role of namespace in XML? Explain with an example.
5. Write a HTML script to create a table with three rows and three columns. The second row and third column should contain a text "TU" with a link set to www.tu.edu.np . The height of the second row should be set to 200px.
6. What is XML? Differentiate between XML schema and DTD.
7. WAP to set and retrieve cookies of a user data username and email.

Group C

Attempt any TWO questions. [2 × 10 = 20]

8. Write an HTML script with CSS for following effects
 - a. Link: After visiting the red color and before visiting the blue color.
 - b. Table with odd rows yellow and even rows red color.
 - c. Text color green, text indentation 2cm, text case uppercase.
9. Write a valid and well-formed xml to store details of visitor list of ABC College with following details and validate record with following information.
 - Records must include details of name, mobile, department, related contact person, date, entry time, exit time and visitor as attributes.
 - Mobile must be 10 digits.
 - Visitor may not have any contact person
 - Records must belong to the Admin, IT, Account and Exam Department.
10. Write a server side script for implementing login. Your program should take username and password as input and redirect to homepage if validated otherwise error "try again".



AMBITION COLLEGE

Mid Baneshwor, Kathmandu, Nepal

Pre-board Examination-2082

Subject :- Statistics I

BCA Third Semester

Time: 2 : 45 hours

Full Marks :- 50

Pass Marks :- 20

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group- B

Attempt any SIX questions [6 * 5 = 30]

2. Analysis the monthly salaries paid to the two group of employee

	A	B
No. of employee	60	40
Average	320	330
S.D	43	50

Find out the combined average of both group?

3. A problem in statistics is given to two students A and B whose chance of solving are $\frac{1}{3}$ and $\frac{1}{4}$ respectively. Find the probability that;
- the problem is solved.
 - Only one of them can solved that problem
 - None of them will solve the problem
 - A solve it, but B cannot
 - Both can solve the problem.

4. From the following data find the missing item, if the mean of the distribution is 115.5

Item	110	112	113	117	-	125	128	130
No. of hours	25	17	13	15	14	8	6	2

5. From a group of 3 Nepalese, 4 Indian and 5 Americans a sub-committee of 4 persons are selected at randomly. Find the probability that the sub-committee will consists i) 3 Nepalese and 1 Indian ii) 1 Nepalese, 1 Indian and 2 American iii) 4 American

6. Compute the karl-pearson correlation coefficient of correlation from the following data.

Price A	25	28	35	20	22	30	31	22
Price B	35	39	48	29	30	38	40	32

Also,

- Calculate its probable error
- Interpret if the value of r is significant or not?

7. Fit the poisson distribution

No. of heads	0	1	2	3	4	5
Observed frequencies	2	10	24	38	18	8

8. Fit the regression equation at $x=65$ by using following data

Marks in statistics	57	58	59	60	61	62	63
Marks in JAVA	77	78	75	78	82	82	79

Group- C

Attempt any two questions [2 * 10 = 20]

9. A sample of 60 cars of two makes P and Q is taken and their average running life in years is recorded as follows:

Life (years)	0-2	2-4	4-6	6-8	8-10
Make P	8	12	22	14	4
Make Q	10	14	19	12	5

Which makes shows greater consistency in performing and why?

10. Let X be a normally distributed variable with mean 30 and standard deviation 4. Then compute the following probability

- $P(X < 34)$
- $P(X > 28)$
- $P(26 < X < 34)$
- 10% of the values are less than the values of X ?

11. The following tables gives the number of refrigerators sold by 4 salesmen in three months may June and July. At 5% level of significance.

Months	Salesmen			
	A	B	C	D
May	50	40	48	39
June	46	48	50	45
July	39	44	40	39

- Is there a significant difference in the sales made by the four salesmen?
- Is there a significant difference in the sales made during different months?

Best of Luck



AMBITION COLLEGE

Mid-Baneshwor, Kathmandu, Nepal

First Term Examination-2082

Subject : OOP in Java (Subjective)

Full Marks :- 50

Time: 3 : 00 hours

Pass Marks :- 20

BCA Third Semester

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group B

Attempt any SIX questions. [6 × 5 = 30]

1. Define Multithreading. What are the benefits of using Multithreading? Explain the life cycle of thread.
2. Why is it important to handle exceptions in Java? Write a program to illustrate the use of exception handling.
3. What is a constructor? Explain types of constructors in java. State the rules for creating a constructor.
4. Define Stream. What are the inbuilt streams available in Java I/O package? Discuss data I/P and O/P streams.
5. Describe the structure of a typical java program. Why do we need the import statement?
6. Define java packages. How do you create packages in Java? Explain in detail with syntax and example.
7. What do you mean by a Java applet? How is an applet different from an application? Discuss the life cycle of an applet.

Group C

Attempt any TWO questions. [2 × 10 = 20]

8.
 - a) Define collection class. Explain different Wrapper classes and associated methods in java.
 - b) Define access modifier. Explain access modifier in java with example.
9. Write a program using swing components to find simple interest. Use text fields for inputs and output. Your program should display the result when the user presses a button.
10.
 - a) Difference between Abstract Class and Interface in Java. Explain with the help of example.
 - b) What is the difference between Container and Component in Swing? Explain the concept of MVC in brief.

Best of Luck



AMBITION COLLEGE

Mid-Baneshwor, Kathmandu, Nepal

Pre-Board Examination-2082

Subject :- Data Structures and Algorithms (Subjective)

BCA Third Semester

Time: 3 : 00 hours

Full Marks :- 50

Pass Marks :- 20

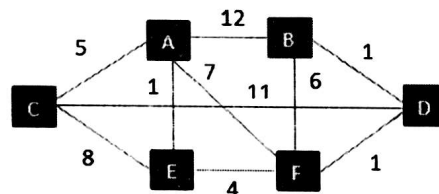
Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group-B

Attempt any two questions

[2 x 10 = 20]

1. Explain linked list. Write the code to create Node class and an algorithm to insert node at the beginning of the linked list. [6+2+2]
2. Discuss about sorting algorithm with an example? Sort the given array using Bubble sort technique: 7, 6, 4, 3. [4+6]
3. How breadth first search and depth first search could be implemented? Explain with an example. Use Dijkstra's algorithm to find the shortest path from vertex A to all other vertices for the following graph. [2+2+6]



Group-C

Attempt any six questions

[6 x 5 = 30]

1. What is Greedy Algorithm? Explain why greedy algorithm doesn't always gives optimal result.
2. What is a heap? Construct a min heap using following values: 27, 7, 10, 35, 12, 17, 11, 3
3. Briefly explain about hashing technique.
4. What is AVL tree? Construct an AVL tree using following data set: 14, 17, 11, 7, 53, 4, 13, 12
5. Explain Huffman algorithm. Encode the given message using Huffman algorithm:

A B B C D B C C D A A B B E E E B E A B E

6. Briefly discuss about divide and conquer algorithms with an example.
7. Explain the complexity of sorting algorithm using linear and binary search.
8. Write short notes (Any Two):
 - i. Big O notation
 - ii. Deterministic and Non-deterministic Algorithm